Part II

Exercises

Chapter 8

Exercise Collection

The following exercises are a selection from the weekly sheets of previous years, arranged by chapter. They range from routine computations to proofs of the kind expected in the exam. Exercises marked with $\star$ are more demanding. Solutions are given at the end of this chapter for exercises marked (S); the remaining exercises are discussed in the plenary exercise session – please do not ask the tutors for written solutions. For some computational exercises a *result for checking* is given.

> **How to write up a proof.** State clearly what you assume and what you want to show. Use every hypothesis (if you did not, either the statement is stronger than it needs to be or your proof is wrong). Justify each step by a definition, a result from the lecture (with number) or a computation. Write in full sentences; "⇒" is not a substitute for an argument.

8.1 Preliminaries and algebraic structures

Exercise 1 (S). Let $f\colon X \to Y$ and $g\colon Y \to Z$ be maps.

(a) Show: if $g \circ f$ is injective, then f is injective.

(b) Show: if $g \circ f$ is surjective, then g is surjective.

(c) Give an example in which $g \circ f$ is bijective, but f is not surjective and g is not injective.

Exercise 2. Let $f\colon X \to Y$ be a map and $A, B \subseteq X$, $C, D \subseteq Y$.

(a) Show that $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$ and $f(A \cap B) \subseteq f(A) \cap f(B)$.

(b) Find an example where the inclusion in (a) is strict, and show that equality holds for all A, B if f is injective.

Exercise 3 (S).

(a) Determine 3^{-1} in $\mathbb{F}_7$ and solve the equation $3x + 4 = 1$ in $\mathbb{F}_7$.

(b) Let K be a field with $\operatorname{char} K = 2$. Show that $(a+b)^2 = a^2 + b^2$ for all $a, b \in K$.

(c) Let v_1, v_2, v_3 be linearly independent vectors in a K-vector space V. Show that $(v_1+v_2,\ v_2+v_3,\ v_3+v_1)$ is linearly independent if $\operatorname{char} K \neq 2$ and linearly dependent if $\operatorname{char} K = 2$.

Exercise 4 ($\star$). Let G be a group in which $g^2 = e$ for all $g \in G$. Show that G is abelian. Find a non-abelian group in which $g^3 = e$ for all g. *(Hint: upper triangular 3×3 matrices over $\mathbb{F}_3$ with ones on the diagonal.)*

Exercise 5. Let $\varphi\colon G \to H$ be a group homomorphism and $K \subseteq H$ a subgroup. Show that $\varphi^{-1}(K)$ is a subgroup of G and that $\varphi(G)$ is a subgroup of H.

8.2 Linear systems

Exercise 6 (S). For which $a \in \mathbb{R}$ is the system

$$x_1 + x_2 + x_3 = 1, \qquad x_1 + a x_2 + x_3 = 2, \qquad x_1 + x_2 + a x_3 = 3$$

solvable? Determine the solution set in each case.

Exercise 7. Solve the following system over $\mathbb{F}_2$ and determine the number of its solutions:

$$x_1 + x_2 + x_4 = 1, \qquad x_2 + x_3 + x_5 = 0, \qquad x_1 + x_3 + x_4 + x_5 = 1.$$

Result for checking: 8 solutions.

8.3 Vector spaces

Exercise 8 (S). Decide which of the following sets are subspaces (with proof or counterexample).

(a) $U_1 = \{\, x \in \mathbb{R}^3 \mid x_1 + x_2 = x_3 \,\} \subseteq \mathbb{R}^3$;

(b) $U_2 = \{\, x \in \mathbb{R}^3 \mid x_1 x_2 = 0 \,\} \subseteq \mathbb{R}^3$;

(c) $U_3 = \{\, f \in \mathbb{R}^{\mathbb{R}} \mid f(1) = 2f(0) \,\} \subseteq \mathbb{R}^{\mathbb{R}}$;

(d) $U_4 = \{\, f \in \mathbb{R}^{\mathbb{R}} \mid f(t) \geq 0 \text{ for all } t \in \mathbb{R} \,\} \subseteq \mathbb{R}^{\mathbb{R}}$.

Exercise 9 (S). Show that the functions sin, cos and exp are linearly independent in the real vector space $\mathbb{R}^{\mathbb{R}}$.

Exercise 10 (S). Let $V = \mathbb{R}[x]_{\leq 3}$ and $U = \{\, p \in V \mid p(1) = 0 \text{ and } p'(1) = 0 \,\}$. Show that U is a subspace and determine a basis and the dimension of U.

Exercise 11. In $\mathbb{R}^4$ let $U = \operatorname{span}\big((1,2,0,1)^{\mathsf{T}}, (2,1,1,0)^{\mathsf{T}}\big)$ and $W = \operatorname{span}\big((1,-1,1,-1)^{\mathsf{T}}, (0,1,0,1)^{\mathsf{T}}\big)$. Determine $\dim(U+W)$ and a basis of $U \cap W$. Verify the dimension formula (Theorem 4.5.4).

Exercise 12 ($\star$). Let $\operatorname{char} K \neq 2$. Show that $K^{n\times n} = \operatorname{Sym}_n \oplus \operatorname{Skew}_n$, where $\operatorname{Sym}_n = \{\, A \mid A^{\mathsf{T}} = A \,\}$ and $\operatorname{Skew}_n = \{\, A \mid A^{\mathsf{T}} = -A \,\}$, and compute the dimensions of both subspaces. What goes wrong if $\operatorname{char} K = 2$?

Exercise 13 ($\star$). Let V be a vector space over an *infinite* field K and $U_1, \dots, U_k \subsetneq V$ proper subspaces. Show that $V \neq U_1 \cup \cdots \cup U_k$. Show by an example that the statement is false over $\mathbb{F}_2$.

8.4 Linear maps and matrices

Exercise 14 (S). Let $f\colon \mathbb{R}^3 \to \mathbb{R}^2$, $f(x) = (x_1 - x_2 + 2x_3,\ 2x_1 + x_2 + x_3)^{\mathsf{T}}$.

(a) Determine the matrix of f with respect to the standard bases, as well as bases of $\ker f$ and $\operatorname{im} f$.

(b) Determine $M_{\mathcal{C}}^{\mathcal{B}}(f)$ for $\mathcal{B} = \big((1,0,0)^{\mathsf{T}}, (1,1,0)^{\mathsf{T}}, (1,1,1)^{\mathsf{T}}\big)$ and $\mathcal{C} = \big((1,1)^{\mathsf{T}}, (1,-1)^{\mathsf{T}}\big)$.

Exercise 15 (S). Let V be a vector space of dimension n and $f \in \operatorname{End}_K(V)$ with $f \circ f = 0$. Show that $\operatorname{im} f \subseteq \ker f$ and $\operatorname{rank} f \leq n/2$. Give, for every even n, an example with $\operatorname{rank} f = n/2$.

Exercise 16. Show that $\operatorname{rank}(A+B) \leq \operatorname{rank} A + \operatorname{rank} B$ and $\operatorname{rank}(AB) \leq \min\{\operatorname{rank} A, \operatorname{rank} B\}$ for all matrices for which these expressions are defined.

Exercise 17. Compute the inverse of $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix} \in \mathbb{R}^{3\times 3}$ by Gaussian elimination.

Result for checking: $A^{-1} = \begin{pmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{pmatrix}$.

Exercise 18 (S, $\star$). Let $\dim V = n < \infty$ and $f \in \mathrm{End}_K(V)$.

(a) Show that $\ker f \subseteq \ker f^2 \subseteq \ker f^3 \subseteq \cdots$.

(b) Show: if $\ker f^k = \ker f^{k+1}$ for some k, then $\ker f^k = \ker f^{k+m}$ for all $m \geq 0$.

(c) Conclude that $\ker f^n = \ker f^{n+1}$.

Exercise 19 ($\star$). Let $V = K[x]_{\leq n}$ and let $a_0, \ldots, a_n \in K$ be pairwise distinct. Show that $\varphi\colon V \to K^{n+1}$, $p \mapsto (p(a_0), \ldots, p(a_n))^\mathsf{T}$ is an isomorphism, *without* using determinants. Deduce the existence and uniqueness of interpolation polynomials.

8.5 Determinants

Exercise 20. Determine the number of inversions and the sign of $\sigma = \left(\begin{smallmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 5 & 1 & 4 & 2 \end{smallmatrix}\right) \in S_5$, and write σ as a product of transpositions.

Exercise 21 (S). For $n \geq 1$ let D_n be the determinant of the $n \times n$ matrix with entries 2 on the diagonal, -1 directly above and below the diagonal, and 0 elsewhere. Show that $D_n = n + 1$.

Exercise 22 ($\star$). Let $A \in \mathbb{Z}^{n\times n}$. Show that A is invertible with $A^{-1} \in \mathbb{Z}^{n\times n}$ if and only if $\det A \in \{1, -1\}$.

8.6 Eigenvalues

Exercise 23 (S). Let $A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix} \in \mathbb{R}^{3\times 3}$. Determine the eigenvalues and eigenspaces of A, a matrix $S \in \mathrm{GL}_3(\mathbb{R})$ such that $S^{-1}AS$ is diagonal, and a closed formula for A^k, $k \in \mathbb{N}$.

Exercise 24. Show by direct computation that every $A \in K^{2\times 2}$ satisfies $A^2 - \mathrm{tr}(A)\, A + \det(A)\, I_2 = 0$ (*Cayley–Hamilton for* $n = 2$). Conclude that $A^{-1} = \frac{1}{\det A}\big(\mathrm{tr}(A) I_2 - A\big)$ if $\det A \neq 0$.

Exercise 25. Show that A and A^T have the same characteristic polynomial, but in general different eigenspaces. Show that AB and BA have the same characteristic polynomial if A is invertible. (*: and also if not.)

Solutions to selected exercises

Solution to Exercise 1. (a) Let $f(x) = f(x')$. Then $(g \circ f)(x) = (g \circ f)(x')$, hence $x = x'$ by injectivity of $g \circ f$. (b) Let $z \in Z$. Since $g \circ f$ is surjective, there is $x \in X$ with $g(f(x)) = z$; so $y := f(x)$ is a preimage of z under g. (c) $X = Z = \{1\}$, $Y = \{1, 2\}$, $f(1) = 1$, $g(1) = g(2) = 1$. Then $g \circ f = \mathrm{id}_{\{1\}}$ is bijective, $2 \notin f(X)$ and $g(1) = g(2)$. □

Solution to Exercise 3. (a) $3 \cdot 5 = 15 = 1$ in $\mathbb{F}_7$, so $3^{-1} = 5$. Then $3x = 1 - 4 = -3 = 4$, hence $x = 5{\cdot}4 = 20 = 6$. Check: $3{\cdot}6{+}4 = 22 = 1$ in $\mathbb{F}_7$. (b) $(a{+}b)^2 = a^2{+}ab{+}ba{+}b^2 = a^2{+}(1{+}1)ab{+}b^2 = a^2{+}b^2$, since $1{+}1 = 0$. (c) Let $\alpha(v_1{+}v_2){+}\beta(v_2{+}v_3){+}\gamma(v_3{+}v_1) = 0$. Then $(\alpha{+}\gamma)v_1{+}(\alpha{+}\beta)v_2{+}(\beta{+}\gamma)v_3 = 0$, so by independence $\alpha + \gamma = \alpha + \beta = \beta + \gamma = 0$. Adding the three equations gives $2(\alpha + \beta + \gamma) = 0$. If $\operatorname{char} K \neq 2$, then $2 \neq 0$ is invertible, so $\alpha{+}\beta{+}\gamma = 0$, and subtracting the equations gives $\beta = \gamma = \alpha = 0$. If $\operatorname{char} K = 2$, then $\alpha = \beta = \gamma = 1$ is a non-trivial solution: $(v_1{+}v_2){+}(v_2{+}v_3){+}(v_3{+}v_1) = 2(v_1{+}v_2{+}v_3) = 0$. □

Solution to Exercise 6. Subtracting the first equation from the other two gives the equivalent system

$$x_1 + x_2 + x_3 = 1, \qquad (a-1)x_2 = 1, \qquad (a-1)x_3 = 2.$$

For $a = 1$ the second equation reads $0 = 1$: no solution. For $a \neq 1$ we obtain the unique solution $x_2 = \frac{1}{a-1}$, $x_3 = \frac{2}{a-1}$, $x_1 = 1 - \frac{3}{a-1} = \frac{a-4}{a-1}$. □

Solution to Exercise 8. (a) Yes: $U_1 = L(A, 0)$ for $A = (1\ 1\ {-1})$, a subspace by Lemma 3.1.2. (b) No: $e_1, e_2 \in U_2$, but $e_1 + e_2 \notin U_2$. (U_2 is the union of two planes.) (c) Yes: the zero function lies in U_3; if $f(1) = 2f(0)$ and $g(1) = 2g(0)$, then $(f + \lambda g)(1) = 2(f + \lambda g)(0)$. (d) No: the constant function 1 lies in U_4, but $(-1) \cdot 1 \notin U_4$. (U_4 is closed under addition and under multiplication by $\lambda \geq 0$; it is a so-called *cone*.) □

Solution to Exercise 9. Let $a\sin{+}b\cos{+}c\exp = 0$, i.e. $a\sin t + b\cos t + ce^t = 0$ for all $t \in \mathbb{R}$. Differentiating twice gives $-a\sin t - b\cos t + ce^t = 0$ for all t. Adding both equations yields $2ce^t = 0$, so $c = 0$. Then $t = 0$ gives $b = 0$, and $t = \pi/2$ gives $a = 0$. (Alternatively, without differentiation: evaluate at sufficiently many points, or compare growth as $t \to \infty$.) □

Solution to Exercise 10. The map $\varphi\colon V \to \mathbb{R}^2$, $p \mapsto (p(1), p'(1))^{\mathsf{T}}$, is linear, and $U = \ker\varphi$ is a subspace. φ is surjective because $\varphi(1) = (1, 0)^{\mathsf{T}}$ and $\varphi(x - 1) = (0, 1)^{\mathsf{T}}$. By Theorem 5.2.1, $\dim U = 4 - 2 = 2$. The polynomials $(x-1)^2$ and $x(x-1)^2$ lie in U (both vanish at 1 together with their derivatives, by the product rule) and are linearly independent, as they have different degrees. Hence $\big((x-1)^2, x(x-1)^2\big)$ is a basis of U by Theorem 4.4.5(b). □

Solution to Exercise 14. (a) $A = \left(\begin{smallmatrix} 1 & -1 & 2 \\ 2 & 1 & 1 \end{smallmatrix}\right)$. Gaussian elimination: $\left(\begin{smallmatrix} 1 & -1 & 2 \\ 0 & 3 & -3 \end{smallmatrix}\right)$, so $x_2 = x_3$ and $x_1 = x_2 - 2x_3 = -x_3$; thus $\ker f = \operatorname{span}\big((-1, 1, 1)^{\mathsf{T}}\big)$. By rank–nullity $\operatorname{rank} f = 2$, so $\operatorname{im} f = \mathbb{R}^2$ with basis (e_1, e_2).

(b) $f(b_1) = (1, 2)^{\mathsf{T}}$, $f(b_2) = (0, 3)^{\mathsf{T}}$, $f(b_3) = (2, 4)^{\mathsf{T}}$. A vector y has $\mathcal{C}$-coordinates $\big(\frac{y_1+y_2}{2}, \frac{y_1-y_2}{2}\big)$, since $p(1, 1)^{\mathsf{T}} + q(1, -1)^{\mathsf{T}} = (p + q, p - q)^{\mathsf{T}}$. Hence

$$M_{\mathcal{C}}^{\mathcal{B}}(f) = \begin{pmatrix} 3/2 & 3/2 & 3 \\ -1/2 & -3/2 & -1 \end{pmatrix}.$$

□

Solution to Exercise 15. If $w = f(v) \in \operatorname{im} f$, then $f(w) = f(f(v)) = 0$, so $w \in \ker f$. Hence $\operatorname{rank} f = \dim\operatorname{im} f \leq \dim\ker f = n - \operatorname{rank} f$ by Theorem 5.2.1, i.e. $\operatorname{rank} f \leq n/2$. For $n = 2m$ and a basis $(v_1, \ldots, v_m, w_1, \ldots, w_m)$ define f by $f(v_i) = w_i$ and $f(w_i) = 0$ (Theorem 5.1.5); then $f \circ f = 0$ and $\operatorname{im} f = \operatorname{span}(w_1, \ldots, w_m)$ has dimension m. In matrix form: $\left(\begin{smallmatrix} 0 & 0 \\ I_m & 0 \end{smallmatrix}\right)$. □

Solution to Exercise 18. (a) $f^k(v) = 0$ implies $f^{k+1}(v) = f(0) = 0$. (b) By induction on m it suffices to show $\ker f^{k+2} \subseteq \ker f^{k+1}$. Let $f^{k+2}(v) = 0$. Then $f^{k+1}(f(v)) = 0$, so $f(v) \in \ker f^{k+1} = \ker f^k$, i.e. $f^{k+1}(v) = f^k(f(v)) = 0$. (c) The dimensions $d_k = \dim\ker f^k$ form an increasing sequence $0 = d_0 \leq d_1 \leq \cdots \leq n$. If $d_0 < d_1 < \cdots < d_{n+1}$, then $d_{n+1} \geq n + 1$, which is impossible. So $d_k = d_{k+1}$, i.e. $\ker f^k = \ker f^{k+1}$ (Theorem 4.4.6), for some $k \leq n$, and (b) gives $\ker f^n = \ker f^{n+1}$. □

Solution to Exercise 21. $D_1 = 2$, $D_2 = 4 - 1 = 3$. For $n \geq 3$, expand along the first row (Theorem 6.4.1): the first term is $2D_{n-1}$; the second is $-(-1)\det M$, where M is the matrix obtained by deleting row 1 and column 2. The first column of M is $(-1, 0, \ldots, 0)^{\mathsf{T}}$, and expanding M along it gives $\det M = -D_{n-2}$. Hence $D_n = 2D_{n-1} - D_{n-2}$. By induction, $D_n = 2n - (n - 1) = n + 1$. □

Solution to Exercise 23. Write $A = I + J$ with J the all-ones matrix. J has rank 1 and $J(1,1,1)^{\mathsf{T}} = 3(1,1,1)^{\mathsf{T}}$. Hence $\operatorname{Eig}(A,1) = \ker J = \{\, x \mid x_1 + x_2 + x_3 = 0 \,\} = \operatorname{span}\big((1,-1,0)^{\mathsf{T}}, (1,0,-1)^{\mathsf{T}}\big)$ and $\operatorname{Eig}(A,4) = \operatorname{span}\big((1,1,1)^{\mathsf{T}}\big)$. As $2+1=3$, A is diagonalisable, and $\chi_A = (x-1)^2(x-4)$. With

$$S = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \\ 0 & -1 & 1 \end{pmatrix} \quad \text{we get} \quad S^{-1}AS = \operatorname{diag}(1,1,4).$$

Since $J^2 = 3J$, we have $J^i = 3^{i-1}J$ for $i \geq 1$, and by the binomial theorem (valid as I and J commute)

$$A^k = \sum_{i=0}^{k} \binom{k}{i} J^i = I + \Big(\sum_{i=1}^{k} \binom{k}{i} 3^{i-1}\Big) J = I + \frac{4^k - 1}{3}\, J.$$

Check: $k = 1$ gives $I + J = A$. □